

Project Name: AI Code Navigator (Backend) (week – 10)

Problem: Slow and Unsafe GitHub Repo Upload in AI Code Navigator

What Was the Problem?

When I uploaded a GitHub repository, the system was processing every single file one by one. It would wait for the first file to finish before starting the second one. This was a problem because:

- For a repo with 50 files, it had to wait for 50 OpenAI API calls one after another which was very slow
- OpenAI has a rate limit if too many requests are sent at once or too fast it returns an error and the whole upload would crash
- If even one file caused an error, the entire repo upload would stop and nothing would be saved

How Did We Fix It?

The idea was simple instead of processing one file at a time, we process small groups of 3 files at a time

Think of it like a checkout counter. Before the fix it was one cashier serving one customer at a time. After the fix it is 3 cashiers serving 3 customers at the same time, then the next 3

We used JavaScript's built-in **Promise.all()** to run multiple files at the same time and a simple helper function called **chunkArray()** to split the file list into groups of 3

We also wrapped each file in its own **try/catch** so if one file fails, it just logs a warning and skips that file instead of crashing the whole upload.

What Files We Changed

1. services/githubService.js

We added a helper function called `chunkArray()` at the top that takes a list and splits it into smaller groups of a given size.

We then rewrote the `processRepo()` function in two steps:

- Step 1 collect all valid files into a list first without doing any API calls
- Step 2 split that list into batches of 3 and process each batch using `Promise.all()`

Each file inside the batch has its own try/catch so one bad file does not stop the rest.

How We Tested It

We tested the /api/upload-repo endpoint with a real GitHub repository:

- Sent a POST request to /api/upload-repo with the repo expressjs/serve-static
- The server processed all 13 files in batches of 3
- All 13 files were saved to MongoDB with their summaries and embeddings
- No errors or crashes during the upload

Response received:

```
{ "success": true, "message": "Repository processed. 13 files saved, 0 duplicates  
skipped.", "totalFiles": 13, "saved": 13, "skipped": 0 }
```

Result

Before the fix, uploading a large repo would either take a very long time or crash halfway because of rate limit errors from OpenAI. If any one file failed, the whole upload was lost

After the fix, files are processed 3 at a time which is faster and stays within OpenAI rate limits. If one file fails it is skipped and the rest continue saving normally